

DBMS – PRODUCT AND CATEGORY MANAGEMENT

Project title- Meesho Jewellery Marketplace Database

1. Database Design

1.1 category table

Attribute	Data Type	Constraint
CATEGORY_ID	NUMBER(5)	Primary Key
CATEGORY_NAME	VARCHAR2(50)	NOT NULL
CATEGORY_DESCRIPTION	VARCHAR2(200)	—

1.2 Product Table

Attribute	Data Type	Constraint
PRODUCT_ID	NUMBER(5)	Primary Key
PRODUCT_NAME	VARCHAR2(100)	NOT NULL
DESCRIPTION	VARCHAR2(200)	—
PRICE	NUMBER(10,2)	NOT NULL
STOCK	NUMBER(5)	NOT NULL
IMAGE	VARCHAR2(200)	—
SELLER_ID	NUMBER(5)	—
CATEGORY_ID	NUMBER(5)	Foreign Key

2. Primary and Foreign Key Relationship

Category → Product

- CATEGORY.CATEGORY_ID → Primary Key
- PRODUCT.PRODUCT_ID → Primary Key
- PRODUCT.CATEGORY_ID → Foreign Key referencing CATEGORY.CATEGORY_ID
- Relationship: One-to-Many (1:M)

3. Table Creation

3.1 Create Category Table

```
SQL> CREATE TABLE CATEGORY (  
2     CATEGORY_ID NUMBER(5) PRIMARY KEY,  
3     CATEGORY_NAME VARCHAR2(50) NOT NULL,  
4     CATEGORY_DESCRIPTION VARCHAR2(200)  
5 );  
  
Table created.
```

3.2 Create Product Table

```
SQL> CREATE TABLE PRODUCT (  
2     PRODUCT_ID NUMBER(5) PRIMARY KEY,  
3     PRODUCT_NAME VARCHAR2(100) NOT NULL,  
4     DESCRIPTION VARCHAR2(200),  
5     PRICE NUMBER(10,2) NOT NULL,  
6     STOCK NUMBER(5) NOT NULL,  
7     IMAGE VARCHAR2(200),  
8     SELLER_ID NUMBER(5),  
9     CATEGORY_ID NUMBER(5),  
10    CONSTRAINT FK_PRODUCT_CATEGORY  
11        FOREIGN KEY (CATEGORY_ID)  
12        REFERENCES CATEGORY(CATEGORY_ID)  
13 );  
  
Table created.
```

4. Data Insertion

4.1 Category Records

```
SQL> INSERT INTO CATEGORY VALUES (1, 'Earrings', 'Different types of jewelry earrings');  
1 row created.  
  
SQL> INSERT INTO CATEGORY VALUES (2, 'Necklaces', 'Necklaces and chains');  
1 row created.  
  
SQL> INSERT INTO CATEGORY VALUES (3, 'Rings', 'Fashion and traditional rings');  
1 row created.  
  
SQL> INSERT INTO CATEGORY VALUES (4, 'Bangles', 'Bangles and bracelets');  
1 row created.  
  
SQL>  
SQL> INSERT INTO CATEGORY VALUES (5, 'Bracelets', 'Fashion bracelets');  
1 row created.
```

4.2 Product Records

```
SQL> INSERT INTO PRODUCT VALUES
  2 (101, 'Gold Plated Earrings', 'Elegant gold plated earrings', 299.00, 20, 'earrings.jpg', 1, 1);

1 row created.

SQL> INSERT INTO PRODUCT VALUES
  2 (102, 'Pearl Necklace', 'Beautiful pearl necklace', 499.00, 15, 'necklace.jpg', 2, 2);

1 row created.

SQL> INSERT INTO PRODUCT VALUES
  2 (103, 'Fashion Ring', 'Stylish fashion ring', 199.00, 25, 'ring.jpg', 3, 3);

1 row created.

SQL> INSERT INTO PRODUCT VALUES
  2 (104, 'Traditional Bangles', 'Traditional decorative bangles', 349.00, 30, 'bangles.jpg', 4, 4);

1 row created.

SQL> INSERT INTO PRODUCT VALUES
  2 (105, 'Charm Bracelet', 'Trendy charm bracelet', 249.00, 18, 'bracelet.jpg', 5, 5);

1 row created.
```

5. Data Retrieval

5.1 Display Category Records

```
SQL> SELECT *FROM CATEGORY;

CATEGORY_ID CATEGORY_NAME
-----
CATEGORY_DESCRIPTION
-----
          1 Earrings
Different types of jewelry earrings

          2 Necklaces
Necklaces and chains

          3 Rings
Fashion and traditional rings

CATEGORY_ID CATEGORY_NAME
-----
CATEGORY_DESCRIPTION
-----
          4 Bangles
Bangles and bracelets

          5 Bracelets
Fashion bracelets
```

5.2 Display Product Record

```
SQL> SELECT * FROM PRODUCT;
```

```
PRODUCT_ID
```

```
PRODUCT_NAME
```

```
DESCRIPTION
```

```
PRICE
```

```
STOCK
```

```
IMAGE
```

```
SELLER_ID CATEGORY_ID
```

```
101
```

```
PRODUCT_ID
```

```
PRODUCT_NAME
```

```
DESCRIPTION
```

```
PRICE
```

```
STOCK
```

```
IMAGE
```

```
SELLER_ID CATEGORY_ID
```

```
Gold Plated Earrings
```

```
PRODUCT_ID
```

```
PRODUCT_NAME
```

```
DESCRIPTION
```

6.Product update

```
SQL> UPDATE PRODUCT
2 SET PRICE = 600.00,
3     STOCK = 50
4 WHERE PRODUCT_ID = 102;
```

```
1 row updated.
```

7.Product Delete

```
SQL> DELETE FROM PRODUCT
2 WHERE PRODUCT_ID = 105;
```

```
1 row deleted.
```

8. Category-wise Product Reports

8.1 Product Listing by Category

This report displays the category name along with the products belonging to each category.

```
SQL> SELECT C.CATEGORY_NAME,  
2         P.PRODUCT_NAME,  
3         P.PRICE,  
4         P.STOCK  
5 FROM CATEGORY C  
6 JOIN PRODUCT P  
7 ON C.CATEGORY_ID = P.CATEGORY_ID  
8 ORDER BY C.CATEGORY_NAME;
```

```
CATEGORY_NAME  
-----  
PRODUCT_NAME  
-----  
PRICE      STOCK  
-----  
Bangles  
Traditional Bangles  
349        30
```

```
Earrings  
Gold Plated Earrings  
299        20
```

```
CATEGORY_NAME  
-----  
PRODUCT_NAME  
-----  
PRICE      STOCK  
-----  
Necklaces  
Pearl Necklace  
600        50
```

```
Rings  
Fashion Ring
```

8.2 Product Listing By Specific Category

```
SQL> SELECT C.CATEGORY_NAME,  
2         P.PRODUCT_NAME,  
3         P.PRICE,  
4         P.STOCK  
5 FROM CATEGORY C  
6 JOIN PRODUCT P  
7 ON C.CATEGORY_ID = P.CATEGORY_ID  
8 WHERE C.CATEGORY_NAME = 'Necklaces';
```

```
CATEGORY_NAME  
-----  
PRODUCT_NAME  
-----  
PRICE      STOCK  
-----  
Necklaces  
Pearl Necklace  
600        50
```

8.3 Product Count by Category

```
SQL> SELECT C.CATEGORY_NAME,  
2         COUNT(P.PRODUCT_ID) AS PRODUCT_COUNT  
3   FROM CATEGORY C  
4  LEFT JOIN PRODUCT P  
5    ON C.CATEGORY_ID = P.CATEGORY_ID  
6  GROUP BY C.CATEGORY_NAME  
7  ORDER BY C.CATEGORY_NAME;
```

CATEGORY_NAME	PRODUCT_COUNT
Bangles	1
Bracelets	0
Earrings	1
Necklaces	1
Rings	1